



Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Introduction to Bayesian Analysis (MTH422A)

Mid-semester examination, Date: Feb 24, 2023, Saturday

Time: 6:00–7:30 PM

Max point: 25

1. If X has support $\mathbb{R}$, for each $q \in \{1, 2\}$, find c such that $\pi(x) = c \exp[-\lambda|x|^q]$ is a valid PDF. Here, λ is positive and finite (known). Now, suppose you have a Bayesian linear regression setting without intercept, i.e., $Y_i|\beta \stackrel{\text{Indep}}{\sim} \text{Normal}(\beta X_i, 1), i = 1, \dots, n$, and you want to use $\pi(\cdot)$ as a prior for β . Show that the MAP estimators of β for $q \in \{1, 2\}$ coincide with LASSO and ridge estimators. (2 + 3 = 5 Points)
2. Suppose $\{X_t\}, t \geq 1$ is Markov time series observed over $t \in \{1, 2, \dots, T\}$. Here, $X_1 \sim \text{Normal}(0, \sigma^2)$ and $X_t|X_{t-1} \sim \text{Normal}(\rho X_{t-1}, (1 - \rho^2)\sigma^2), t = 2, \dots, T$. The autocorrelation parameter ρ is known and $\rho \in (-1, 1)$. We choose a prior $\sigma^2 \sim \text{IG}(0.1, 0.1)$. Find the posterior mean and variance of σ^2 . Now, suppose you want to implement this Bayesian learning in an online set-up where data are flowing into the system sequentially and say, $T \approx 10^9$. Once X_{T+1} observed, how would you efficiently update the posterior summaries? No point for an inefficient approach. Assume that you do not have any multi-core system. (2 + 2 = 4 Points)
3. Suppose $\{\mathbf{X}_t\} \in \mathbb{R}^P, t \geq 1$ is a multivariate continuous-valued Markov time series observed over the time points $t \in \{1, 2, \dots, T\}$. Suppose, $\mathbf{X}_1 \sim \text{Normal}_P(\mathbf{0}, \Sigma)$, and $\mathbf{X}_t|\mathbf{X}_{t-1} \sim \text{Normal}_P(\rho \mathbf{X}_{t-1}, \Sigma), t = 2, \dots, T$. The autocorrelation parameter is $\rho \in (-1, 1)$. Assume that ρ is known and Σ is unknown. Assuming a prior $\Sigma \sim \text{Inverse-Wishart}(\mathbf{I}_P, \nu)$, calculate the posterior mean (analytical expression) of Σ . The notations are the same as in the lecture notes. Here $\mathbf{I}_P$ denotes the identity matrix of dimension $P \times P$. (2 Points)
4. Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$. First derive the Jeffreys' prior for λ . Then, using it, obtain the posterior distribution of λ given $X_1, \dots, X_n$. Find the mean and variance of the MAP estimator of λ . Obtain the posterior predictive distribution $\pi(X_{n+1}|X_1, \dots, X_n)$. (2+1+3+2 = 8 Points).
5. In each of the last two seasons, India played 10 cricket matches. In the first season, India won 4 matches, lost 4 matches, and 2 matches were draw. In the second season, India won 5 matches, lost 3 matches, and 2 matches were draw. Let $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)'$ be the vector of your win, lose, and draw probabilities. Assume a uniform prior distribution for $\boldsymbol{\theta}$ over its support. If India plays 10 games in each of the next three seasons (with the same difficulty level of the competitors), what is the posterior predictive probability that India win 9 and lose 1 in the next season? What is the posterior predictive probability that India will win 8, lose 1, and draw 1 in the **next-to-next** season? What is the posterior predictive probability that India will win or draw in 8, and lose in 2 in the **next-to-next-to-next** season? Final answer must be in the form of gamma functions only and you do not need to evaluate them using a calculator. (2 + 2 + 2 = 6 Points)

Results: If $\mathbf{X} = (X_1, \dots, X_P) \sim \text{Multinomial}(N; \theta_1, \dots, \theta_P)$, then $\sum_{i \in \mathcal{A}} X_i \sim \text{Binomial}(N, \sum_{i \in \mathcal{A}} \theta_i)$, where $\mathcal{A}$ is a subset of the indexes $\{1, \dots, P\}$. If $\mathbf{X} = (X_1, \dots, X_P) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_P)$, then $\sum_{i \in \mathcal{A}} X_i \sim \text{Beta}(\sum_{i \in \mathcal{A}} \alpha_i, \sum_{i \in \mathcal{A}^c} \alpha_i)$, where $\mathcal{A}^c$ denotes the complement of $\mathcal{A}$.